

Reasoning or Pattern Matching? Multi-Metric and External Judge Evaluation of LLMs on Bayesian Statistical Reasoning

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Abstract—Accuracy-only leaderboards disguise the actual failure modes of LLMs when used for solving statistical problems. Five frontier models — Claude Sonnet 4.5, GPT-4.1, Gemini 2.5 Flash, DeepSeek-Chat, Mistral Large — are benchmarked on 171 Bayesian and inferential statistics tasks across 38 topics. Furthermore, three axes perturbations — rephrasing, numerical change, and semantic change — are also applied, summing up to a total of 473. Each response is scored twice, by a Python-based deterministic keyword rubric and by an out-of-family independent AI judge — Llama 3.3 70B Turbo via Together AI on assumption articulation and reasoning quality. Three leaderboards are derived from the results. GPT-4.1 is first on robustness to prompt variation (0.9996), but ranks third on NMACR accuracy (0.673). Claude leads in verbalized extraction of self-confidence (ECE 0.033) and places second on NMACR accuracy (0.698), but drops to fourth in robustness to prompt variation (0.9565). 44.3% of computed 750 paired (task, response) outputs fail on the assumption-compliance dimension (no mention of assumption derivation). Among 143 judge-flagged Level-1 (assumption-compliance) failures, 67 (46.9%) are assumption violations. There is a moderate agreement on assumption compliance between the keyword rubric (regex from Python-based ground-truth) and the Llama judge, with a Spearman $\rho = 0.602$ on assumption compliance, essentially zero ($\rho = 0.004$) on method structure, and weak agreement ($\rho = 0.180$) on reasoning quality. A single-metric leaderboard is the wrong approach in such evaluations, so a deeper and multi-dimensional assessment will serve as a stronger foundation.

Index Terms—LLMs, Bayes’ theorem, calibration, perturbation robustness, expected calibration error, LLM-as-judge, benchmark

I. INTRODUCTION

Increased use of large language models as informal statistical assistants is evident in the fact that most students reach for them for posterior derivation, practitioners are inquiring sketches for hierarchical models, and reviewers consult with them on the assumption articulation behind the testing. Whether they are fit for these roles is discussed and answered in single-accuracy metric leaderboards across dozens of research online. Single metric evaluation is convenient, but it fails in the case of three properties that are of great concern to a careful user, who prefers separate explanations for them. The first issue is whether the model produces the correct value

when there is a clean problem statement. The second issue is that after rephrasing, the model continues to derive the correct value, even when the problem is recast into a different domain, or is embodied with new inputs. The third issue is whether the empirical correctness of the model corresponds to its stated confidence. Prior LLM-statistics work informs this evaluation pipeline differently along each axis. Accuracy-focused benchmarks like MathEval [1] and the LLM-vs-human study of Nagarkar et al. [2] score the first property exclusively. StatEval [3] adds proof-evaluation rubrics but does not measure robustness or calibration. Longjohn et al. [4] treat the evaluation itself probabilistically rather than the reasoning content. A single-column leaderboard still picks a winner, even when there is a strong disagreement on the three properties of the same set of models.

The novelty and contribution of this work is an evaluation pipeline that is designed to independently measure and compare the three properties. The starting point of the pipeline is a task set of 171 Bayesian and inferential problems across 38 topics, covering both closed-form conjugate inference and computational methods, such as Gibbs sampling, Metropolis-Hastings, Hamiltonian Monte Carlo, Reversible-Jump MCMC, hierarchical models, Variational Bayes, and Approximate Bayesian Computation. Based on the standard textbooks [5], [6], [7], each task is paired with a ground-truth solver, and the solutions are not visible or accessible to LLMs in any form. The perturbation of a task is done along three axes: rephrasing (same math, reworded prompt), semantic (same math, different real-world context), and numerical (changed math inputs with recomputed ground-truth), summing up to total 473 perturbation records. Both the base tasks and perturbations are simultaneously dispatched to five models (Claude Sonnet 4.5, GPT-4.1, Gemini 2.5 Flash, DeepSeek-Chat, and Mistral Large) through a unified HTTP client, which takes records of latency, token counts, and raw output. Scoring of each response is done along three paths: a deterministic regex extractor for the final numeric answer, a five-dimensional NMACR rubric over Numeric Accuracy, Method Structure, Assumption Compliance, Confidence Calibration, and Reason-

ing Quality, and an out-of-family (totally independent from the five LLMs) judge — Llama 3.3 70B Turbo, via Together AI — which re-scores the three qualitative dimensions. The aforementioned scoring paths are fed to three separate aggregations, which then produce three rankings instead of one. The single-judge protocol has become the default in the literature, so the dual scoring step deserves a brief justification. Despite being cheap and reproducible, the keyword rubric confuses surface with core substance. That is, a response containing the word “assume” does not necessarily categorize as a response with a usable assumption. The retrieved baseline information [8], [9] states that a single LLM judge can read the substance, but exhibits self-preference and design-choice sensitivity. To convert that methodological tension into a measurable quantity, both graders are run on each response. The score is defensible when the two graders agree, that is, give a PASS based on their thresholds. The disagreement gap itself is recorded as a finding rather than whitewashed and treated as evidence that there is a high chance of overestimation of Bayesian competence in the keyword-only protocols used earlier [3], [1].

Contributions:

- 1) 171 Bayesian and inferential tasks are benchmarked (136 closed-form conjugate models, 35 computational methods including Gibbs, Metropolis-Hastings, HMC, RJMCMC, hierarchical, VB, ABC), with three types of perturbations for all 38 task types.
- 2) Deterministic keyword rubric is cross-evaluated with independent and open weights judge — Llama 3.3 70B Turbo. Agreement rates are reported per dimension and the disagreement gap between the rubric and the judge is itself a contributing finding.
- 3) Empirical evidence that the accuracy, robustness, and calibration rankings of the same five LLMs disagree essentially. On a single-metric leaderboard, neither aforementioned pattern appears.
- 4) Documented and reproducible methodology. NMACR weights ($A=0.30$, $R=0.25$, $M=0.20$, $C=0.15$, $N=0.10$), aggregation paths, and tolerance thresholds are specified inline in §III and motivated from the cited literature.

II. LITERATURE REVIEW

Quantitative-reasoning benchmarks. StatEval [3] contains 13,817 problems across several dimensions, including descriptive statistics, probability, inference, and regression, and a mix of answer correctness and rubric-based proof evaluation serves as a scoring method. MathEval [1] compares 52 LLMs across 22 mathematical-reasoning datasets, again on accuracy. Nagarkar et al. [2] compare the responses of LLMs to humans on statistical tasks. All three mainly take into account how often the model derives the right number. None of these papers asks whether the reasoning is well-formed or stable under perturbation.

Bayesian reasoning. Longjohn et al. [4] aggregate benchmark results probabilistically rather than evaluating Bayesian reasoning itself. Du et al. [10] document assumption-blindness in causal reasoning: LLMs produce confident causal claims

that semantic cues drive even when the data contradict them. The pattern this work documents in inferential statistics matches that finding.

Calibration. Guo et al. [11] introduced Expected Calibration Error and showed that modern neural networks, despite being high on accuracy, are, in fact, systematically miscalibrated. For classifiers, ECE is well studied. Applying it to free-form LLM reasoning is recent and less settled. FermiEval and related work [12], [13], [2] usually adapt calibration to natural-language outputs on a single dimension. Binary-pass ECE and a per-NMACR-dimension ECE are reported in this research.

LLM-as-judge. Zheng et al. [14] introduced LLM as a judge with MT-Bench, showing that strong LLMs can substitute for human raters on open-ended tasks. Follow-up work documents design-choice sensitivity and rubric-adherence failures [8], [9]. Brittlebench [15] documents prompt sensitivity. ReasonBench [16] documents run-to-run instability from stochastic decoding. The protocol used here constrains the judge to a single dimension (assumption-compliance) and uses an out-of-family open-weights model (Llama 3.3 70B Turbo). Feuer et al. [9] address both choices of failure modes.

No prior benchmark is known for Bayesian or inferential statistical reasoning that combines (i) a perturbation taxonomy for robustness, (ii) an external single-dimension judge for assumption-compliance, and (iii) calibration measured on the same task set. The contribution of this work is the combination of those three, which surfaces ranking divergences that a single-metric leaderboard cannot.

III. METHODOLOGY

A. Task Suite

The 171 base tasks span two phases across 38 task types. Phase 1 (136 tasks, 31 types) covers closed-form conjugate Bayesian models: Beta-Binomial, Gamma-Poisson, Normal mean with known and unknown variance (Normal-Inverse-Gamma), Dirichlet-Multinomial, Bayesian linear regression, and decision theory under squared, absolute, and 0-1 loss. Phase 2 (35 tasks, 7 types, 5 tasks each) covers computational methods: Gibbs sampling, Metropolis-Hastings, Hamiltonian Monte Carlo, Reversible-Jump MCMC, hierarchical models, Variational Bayes, and Approximate Bayesian Computation. Based on the implementation from textbooks [5], [6], [7], each task has a ground-truth solver. Seeded Monte Carlo estimates are used as ground truth for Phase 2 stochastic methods, with tolerances of 0.05 for MCMC variants and 0.10 for VB and ABC.

B. Perturbation Taxonomy

The full evaluation flow is summarized in Figure 1 below. Task generation feeds into perturbation generation, parallel model querying, and three-path scoring. Perturbation for each task is done in three axes. *Rephrasing* keeps the math and answer while rewording the prompt. *Semantic* perturbation keeps the math, but relocates the problem context to a different real-world framing. *Numerical* perturbation changes the inputs

and recomputes the ground-truth. The full perturbation set is 473 records, 75 of which are hand-written seeds covering 25 base tasks on all three axes. The remaining 398 are LLM-generated perturbations (produced by the LLM judge Llama 3.3 70B Turbo) covering the remaining 146 tasks. Ground-truth solver was checked against each generated perturbation before inclusion. For 40 conceptual or seeded-MC tasks, the numerical axis is skipped because number swapping would cause an invalid ground-truth procedure.

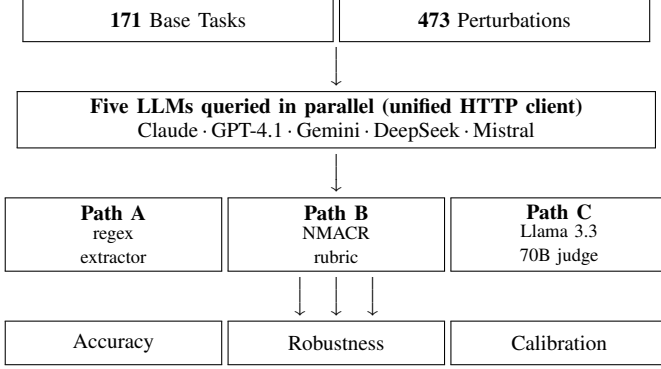


Fig. 1. Evaluation pipeline. Each task and its perturbations are dispatched in parallel to five LLMs via a unified HTTP client. Each response is scored three ways: a deterministic regex extractor (Path A), the NMACR five-dimensional rubric (Path B), and an out-of-family Llama 3.3 70B judge re-scoring the three qualitative dimensions (Path C). The three scoring paths feed three independent aggregations and produce three rankings rather than one.

C. Three-Path Scoring

Path A — the deterministic keyword rubric extracts the answer line, compares numeric targets to ground-truth within a per-task tolerance, and checks required structural and assumption articulation related keywords in the prose. **Path B** — a post-hoc NMACR aggregator. Each task contributes five components from five corresponding domains in the range of $[0, 1]$ — Numerical Accuracy (N), Method Structure (M), Assumption Compliance (A), Confidence Calibration (C), and Reasoning Quality (R). The combined score is

$$S_{\text{NMACR}} = 0.10 N + 0.20 M + 0.30 A + 0.15 C + 0.25 R, \quad (1)$$

with weights drawn from the chain-of-thought, calibration, and assumption-articulation literature [17], [18], [11], [10], [8], [2], [12], [1]. The highest weight in the formula is allocated to A because assumption articulation is the foundation of Bayesian reasoning. N is given the lowest weight since correct numeric calculation without derivation does not support the answer logic of the specified question. The pass threshold is set to $s \geq 0.5$. **Path C** — the external judge runs on one dimension. Llama 3.3 70B Turbo via Together AI is given each (task, response) pair and a fixed prompt rubric, which requires a statement from the judge whether the explicit assumption articulation requirement is met. The holistic-judge failure modes are addressed in a narrower scope in the Feuer et al. [9] paper.

D. Three-Ranking Framework

Three rank orders come from the same response corpus. The *accuracy ranking* uses the mean Path B NMACR score per model on the 171 base tasks. The *robustness ranking* uses the ratio of mean NMACR score on perturbation runs to mean NMACR score on the paired base runs over the $n = 398$ LLM-generated v2 perturbation records (covering 146 of the 171 base tasks); the 75 hand-authored v1 seeds (25 base tasks) are excluded to keep the generation source uniform. With $\bar{s}_{\text{base}}(m)$ and $\bar{s}_{\text{pert}}(m)$ those means,

$$R_m = \frac{\bar{s}_{\text{pert}}(m)}{\bar{s}_{\text{base}}(m)} = 1 - \frac{\Delta_m}{\bar{s}_{\text{base}}(m)}, \quad (2)$$

where $\Delta_m = \bar{s}_{\text{base}}(m) - \bar{s}_{\text{pert}}(m)$ is the signed mean score gap; $R_m = 1$ is perfect agreement. The *calibration ranking* uses binary-pass Expected Calibration Error [11] over B confidence buckets,

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|, \quad (3)$$

with $B = 4$ buckets (low, medium, high, unstated) parsed from the response’s hedging language. The three rankings are not redundant: a model can be accurate but brittle, accurate and well-calibrated but brittle, or robust but inaccurate. §IV shows all three configurations occur in the five-model sample.

E. Experimental Setup

The five LLMs are queried directly over their HTTP APIs with no vendor SDKs: Claude Sonnet 4.5 (Anthropic Messages), GPT-4.1 (OpenAI Chat Completions), Gemini 2.5 Flash (Google Generative Language), DeepSeek-Chat, and Mistral Large. The benchmark run sends no explicit temperature, so each provider’s API default applies (1.0 for Anthropic, OpenAI, Google, and DeepSeek, and 0.7 for Mistral). Max output is 1024 tokens for Claude, GPT-4.1, DeepSeek, and Mistral, and 4096 tokens for Gemini (its thinking-token accounting consumes part of the budget before emitting visible text). Other decoding parameters are shared: a 60-second per-request timeout, retries on transient 429 and 5xx responses, and the same Bayesian-statistics system prompt for every model. Each (task, model) pair produces one response. The calibration analysis uses a separate 3-rerun self-consistency protocol at temperature $T = 0.7$ with top- $p = 0.95$. Bootstrap CIs use 10,000 non-parametric resamples (seed 42) of per-run aggregate scores. A single shell script drives deduplication, scoring, perturbation scoring, judge invocation, and aggregation. 53 unit tests cover the frequentist statistical solvers and the MCP query interface.

TABLE I
PER-MODEL PERFORMANCE ACROSS THE THREE METRICS

Model	Accuracy $\uparrow$	Robustness $\uparrow$	ECE $\downarrow$
Gemini 2.5 Flash	0.731	0.9824	0.0765
Claude Sonnet 4.5	0.698	0.9565	0.0334
GPT-4.1	0.673	0.9996	0.0339
DeepSeek-Chat	0.669	0.9422	0.1977
Mistral Large	0.668	0.9981	0.0811

F. Reproducibility and Audit Infrastructure

Contribution 4 (§I) is reified in the project repository: `reproduce.sh` runs the full pipeline deterministically from raw task prompts through scoring, judge invocation, and aggregation; dependencies are pinned in `requirements.txt` and `environment.yml`; every random seed (e.g., `seed=42` for the 10,000-resample bootstrap) is fixed inline; 53 unit tests cover the frequentist solvers (24 in `test_frequentist.py`) and the MCP query interface (29 in `test_server.py`); and three audit notes in `docs/audit/` (`aggregation_locus.md`, `methodology_continuity.md`, `limitations_disclosures.md`) document the NMACR-weight literature trail, the methodology continuity log, and explicit bias disclosures complementing §VI’s threats to validity. An independent re-runner reaches the same numbers reported in §IV and §V from a fresh clone of the public repository at <https://github.com/albertsimonyan74/bayes-benchmark>.

IV. RESULTS

A. Per-Metric Rankings

The per-model summary appears in Table I. Accuracy is the mean Path B NMACR score on the 171 base tasks. Robustness R_m is the ratio of mean NMACR score on the $n = 398$ paired perturbation runs to mean NMACR score on their base counterparts (Eq. 2), so $R_m = 1$ corresponds to perfect base-to-perturbation agreement. ECE is binary-pass Expected Calibration Error (lower is better).

No two rankings agree. Orderings, rank-flips, and 95% bootstrap CIs are detailed in §IV-B below. DeepSeek is the only model in the bottom two on every axis.

B. Headline Divergence

The cross-metric divergence is concrete: GPT-4.1 ranks third on accuracy (0.673, behind Gemini 0.731 and Claude 0.698) yet its base and perturbation mean NMACR scores fall within 0.0003 (0.6661 vs 0.6658) — first on robustness (0.9996). Claude inverts: second on accuracy (0.698), tied with GPT-4.1 on calibration (ECE $\approx$ 0.033), but mean NMACR drops from 0.7011 to 0.6706 under perturbation, dropping it to fourth on robustness (0.9565).

C. Assumption-Compliance Results

On $n = 750$ paired runs (150 base tasks $\times$ 5 models, excluding 21 task specifications with no required assumption checks; distinct from the 40-task numerical-axis filter in §III) the Llama 3.3 70B Turbo judge passes only 55.7% of responses on assumption-compliance, leaving 44.3% that fail

to articulate the assumptions the derivation depends on, even when the final number is right. The keyword rubric passes 71.1% of the same responses, 15.4 points higher. Spearman rank correlation between keyword and judge scores on this dimension is $\rho = 0.602$; Krippendorff $\alpha = 0.573$ (95% CI [0.516, 0.622], 1,000-resample bootstrap; Table II).

TABLE II
KRIPPENDORFF’S α ACROSS THREE JUDGED DIMENSIONS

Dimension	α	95% CI	Band
Method Structure	−0.009	[−0.072, 0.062]	near chance
Assumption Compliance	0.573	[0.516, 0.622]	questionable
Reasoning Quality	−0.125	[−0.197, −0.059]	near/below chance

Note: Computed on $n = 750$ paired outputs (1,000-resample bootstrap CI). Assumption-compliance $\alpha = 0.573$ is “questionable” ($\alpha < 0.667$); the other two cluster near chance. Per-dimension Spearman correlations: method structure $\rho = 0.004$, reasoning quality $\rho = 0.180$, assumption compliance $\rho = 0.602$. The three-ranking divergence is a within-model cross-metric finding independent of inter-rater agreement.

On the combined base-plus-perturbation pool ($n = 2,850$ = 750 base + 2,100 perturbation rows, eligibility scoped to tasks with at least one required assumption check), 20.7% of responses pass-flip between the keyword rubric and the judge: one in five times, the two scorers disagree on whether the assumption was actually articulated. That is the clearest evidence in the paper for the title’s question. Models produce surface-correct outputs that pass keyword checks for “assume” or “given” tokens while failing the substantive check. Du et al. [10] report the same blindness in causal reasoning.

Figure 2 visualizes the direction of the disagreement. Of 722 disagreeing runs in the $n = 2,850$ pool, 591 are keyword-PASS / judge-FAIL versus only 131 the reverse — a $4.5\times$ asymmetry showing the rubric over-credits surface tokens like “assume” or “given” without substantive assumption articulation.

Keyword Rubric	PASS	1,345 47.2%	591 20.7%
	FAIL	131 4.6%	783 27.5%
		PASS	FAIL
		Judge	

Fig. 2. Keyword-vs-judge disagreement matrix on $n = 2,850$ eligible runs. Keyword-PASS / judge-FAIL outnumbers keyword-FAIL / judge-PASS by $4.5\times$ (591 vs 131): the rubric over-credits surface tokens on assumption-compliance.

D. Failure Taxonomy

The judge tags each failing response with a Level-1 (L1) severity flag when the response fails substantive assumption-compliance independently of whether the final numeric answer

is correct. The 143 L1 responses cluster into five categories. Figure 3 breaks them down per model.

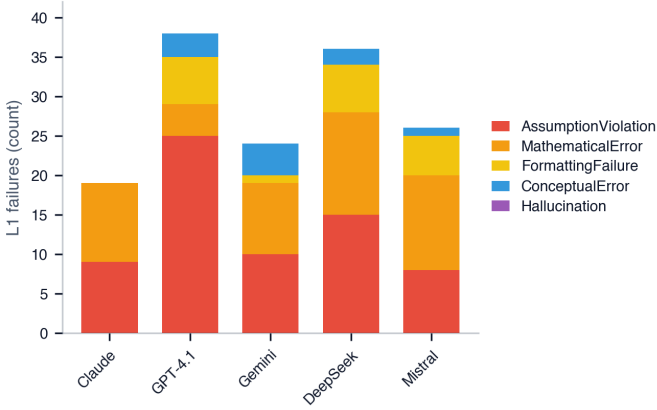


Fig. 3. Failure-mode distribution across the five models ($n = 143$ judge-classified L1 failures, stacked by category). Assumption violations dominate at 46.9% (67), then mathematical errors 33.6% (48), formatting 12.6% (18), and conceptual 7.0% (10). **No hallucinations observed** on these tasks.

Three observations follow. Assumption violations dominate, reinforcing the keyword-vs-judge gap in §IV-C: in Bayesian reasoning the substantive failure mode is unstated or implicit assumptions, not incorrect arithmetic. Hallucinations are absent — fabrication is widely cited as a general LLM risk, but on structured tasks with explicit ground-truth anchors it does not appear here. Third, the per-model breakdown sharpens the GPT-4.1 robustness paradox: GPT-4.1 has the most overall L1 failures (38 of 143), of which 25 are assumption violations. The model ranking first on robustness also omits assumption articulation most; §IV-F1 decomposes this paradox further and shows the per-model picture is more nuanced than absence-of-reasoning-to-vary alone.

E. Calibration Across Extraction Methods

A single ECE number hides where the (mis)calibration comes from. Two extraction methods are separated below. *Verbalized confidence* uses what the model states alongside its answer. *Self-consistency confidence* is inferred from how often the model returns the same answer across reruns at temperature $T = 0.7$. The two measure related but distinct notions: verbalized confidence is a single-pass stated belief the user actually sees; self-consistency is a behavioral probe of whether the model would re-derive the same answer if asked again. A model calibrated only verbally may state “90% confidence” while producing different answers across reruns at the same temperature — pattern-matching on calibration-shaped language rather than tracking actual response stability. The Expected Calibration Error framework of Guo et al. [11] was originally developed for classifiers with explicit logit-derived probabilities; free-form LLM reasoning has no logit anchor, so calibration must be inferred indirectly. The choice between verbal-hedge extraction and behavioral-consistency extraction is itself an unsettled methodological question that this paper exposes by reporting both rather than picking one. The 3-rerun

protocol at $T = 0.7$ balances cost against statistical power: three samples can distinguish an 80% answer-mode rate from a 95% one, the practical regime for frontier LLMs on these tasks.

TABLE III
CALIBRATION ECE BY TWO EXTRACTION METHODS

Model	Verbalized ECE ↓ $n = 855$ base runs	Self-consistency ECE ↓ $n = 2,415$ reruns, $T = 0.7$
Claude	0.033	0.734
GPT-4.1	0.034	0.721
Gemini	0.077	0.618
Mistral	0.081	0.663
DeepSeek	0.198	0.726

Note: Left column = verbalized ECE on $n = 855$ truly-base runs (all 171 base tasks $\times$ 5 models; calibration uses the full base pool with no assumption-check filter, distinct from the $n = 750$ pool in §IV-C). Right column = self-consistency ECE on $n = 2,415$ reruns (3-rerun protocol at $T = 0.7$). All five models exceed ECE = 0.6 on self-consistency, indicating severe overconfidence; the gap between the two columns is direct evidence that stated confidence does not reflect actual response stability.

Table III reframes the story. Verbalized confidence is what a user sees: Claude and GPT-4.1 look well-calibrated, matching the per-model summary. Self-consistency asks instead how often the model agrees with itself across reruns, and the picture inverts: every model crosses the conventional ECE = 0.6 “severe overconfidence” threshold and the cross-model spread collapses. A model that states 90% confidence but reproduces its own answer only 60% of the time is calibrated only verbally. Verbalized confidence is pattern matching on calibration-shaped language, while self-consistency probes underlying stability, and the two come apart by an order of magnitude. The self-consistency ECE treats single-answer agreement across reruns as ground truth, excluding cases where all three reruns fail to produce a parseable answer ($n = 2,415$).

F. Extended Diagnostics: Where the Divergence Lives

The §IV-B headline divergence and the §IV-C/§IV-D failure analyses are reported as model-level summaries. The following subsections decompose those summaries to reveal where the divergence concentrates: which models contribute disproportionately to the keyword-judge disagreement, which task types absorb the failures, which perturbation axes drive the robustness gap, and how the closed-form and computational phases differ.

1) *Per-Model Disagreement Profiles:* The pooled $4.5\times$ asymmetry in §IV-C (Table II) decomposes unevenly across the five models. Mistral has the most lopsided profile: 109 keyword-PASS / judge-FAIL versus only 9 in the reverse direction, a $12.11\times$ asymmetry. Claude is second at $7.84\times$ (149 vs 19), followed by Gemini ($4.19\times$, 130 vs 31) and DeepSeek ($3.12\times$, 100 vs 32). GPT-4.1 is the most balanced at $2.58\times$ (103 vs 40); its keyword passes are also its judge passes more often than any other model in the sample. The over-credit problem is not uniform: two of the five models account for 43.7% of the 591 keyword-PASS / judge-FAIL cases.

This sharpens the GPT-4.1 robustness paradox from §IV-D in an unexpected direction. GPT-4.1 has the most overall L1 failures (38 of 143) yet the lowest keyword-judge asymmetry; its surface tokens track the judge’s substantive read more closely than any other model. The first-place robustness rank therefore does not reduce to pattern-matching of assumption-shaped language — the candidate that did would be Mistral or Claude. GPT-4.1’s robustness instead coincides with substantively correctable failures: when the keyword rubric passes it, the judge agrees; when it fails, both agree. Robustness here reflects within-prompt consistency rather than over-credited surface form.

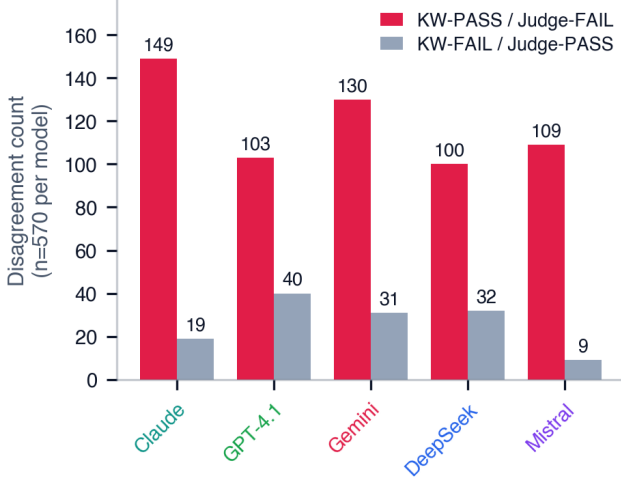


Fig. 4. Per-model keyword-vs-judge disagreement on the $n = 2,850$ eligible pool (570 runs per model). KW-PASS / Judge-FAIL (over-credit) versus KW-FAIL / Judge-PASS (under-credit). The pooled $4.5\times$ asymmetry decomposes unevenly: Mistral $12.11\times$, Claude $7.84\times$, Gemini $4.19\times$, DeepSeek $3.12\times$, GPT-4.1 $2.58\times$.

2) *Failure Concentration by Task Type*: The 143 L1 failures distribute across phases with a counter-intuitive shape: Phase 1 closed-form tasks absorb all 143 (21.03% of 680 Phase 1 responses) while Phase 2 computational methods (Gibbs, MH, HMC, RJMCMC, hierarchical, VB, ABC) absorb zero (0.00% of 175 responses). Computational sampling is not the failure mode the benchmarked LLMs hit at the L1 severity tier; closed-form algebra is. The likely mechanism: Phase 2 prompts invite procedural answers (“run Gibbs for N iterations”) whose ground-truth tolerance is 0.05–0.10 and whose assumption articulation is implicit in the sampler choice, while Phase 1 prompts require explicit prior, likelihood, and posterior derivation that the judge can audit line-by-line.

The 143 failures cluster in 18 of 38 task types; 20 task types are failure-free. The top three are REGRESSION (25), BAYES_REG (25), and BOX_MULLER (19). The first two are particularly stark: every model fails on every problem of those types ($5 \times 5 = 25$ each). This is a TASK-difficulty pattern, not a model-shortcut pattern.

This complicates the §IV-F1 story rather than reinforcing it. The pattern-matching champions identified there (Mistral, Claude) do not dominate the task-type failures: REGRES-

SION and BAYES_REG hit all five models equally, and the bottom of the top-10 (MSE_COMPARE, BAYES_FACTOR, LOG_ML) is dominated by GPT-4.1 and DeepSeek — the models with the lowest §IV-F1 asymmetry. Per-model over-credit on assumption-compliance and per-task-type total L1 failure rate are independent signals.

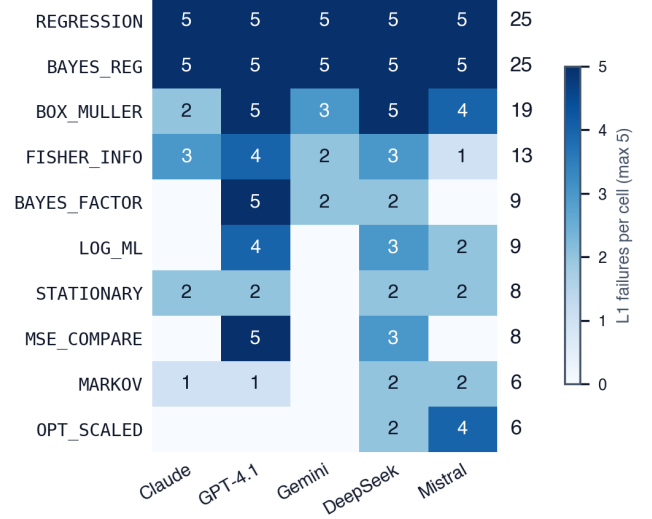


Fig. 5. Top-10 task types by L1 failure count from the $n = 143$ pool, decomposed per model (cell value = failures for that (task type, model) cell; right-margin number = task type total). All ten are Phase 1 closed-form tasks; Phase 2 computational types contribute zero L1 failures. The failures concentrate in 18 of 38 task types (20 types failure-free).

3) *Robustness by Perturbation Axis*: The per-axis pooled means across models invert the prior hypothesis: rephrasing is the most damaging axis at $R = 0.966$, with semantic $R = 0.984$ and numerical $R = 0.978$ slightly less damaging. Rewording prompts hurts more than changing the numbers, indicating sensitivity to surface phrasing even when the underlying math is identical. Numerical perturbation runs only $n = 530$ (40 conceptual or seeded-MC tasks skip that axis per §III) but the spread is the widest: 0.9146 (DeepSeek) to 1.0130 (Mistral), a 0.10-point gap.

Variance decomposition shows robustness is more a model property than an axis property: mean model-variance within axis is 7.28×10^{-4} , mean axis-variance within model is 1.99×10^{-4} — a $3.66\times$ ratio. Table I’s robustness column captures a real model signal, not an average over heterogeneous axis behavior.

Four (model, axis) cells exceed $R = 1.0$, meaning the model scores higher on perturbed prompts than on their base counterparts: GPT-4.1 (semantic 1.011, numerical 1.013) and Mistral (semantic 1.006, numerical 1.013). GPT-4.1’s pooled 0.9996 is therefore not uniform stability but a near-cancellation: a rephrasing drop of 0.021 offsets the two axes where it gains. Cross-linking §IV-F1/§IV-F2: the over-credit ranking (Mistral, Claude) does not align with the robustness ranking (GPT-4.1, Mistral). Mistral is both the largest over-credit contributor and the second-most robust — pattern matching on surface tokens

can be stable across paraphrases even when it is substantively wrong.

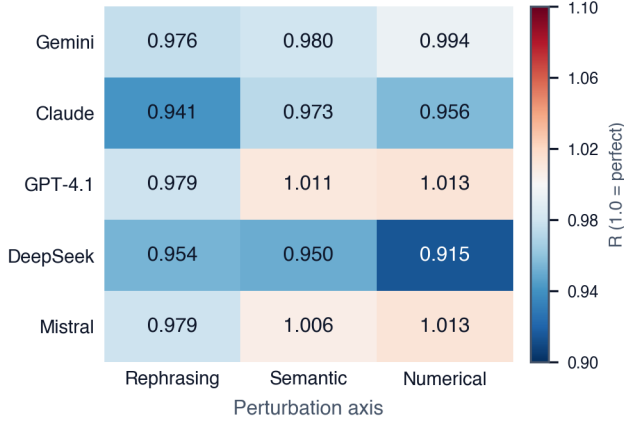


Fig. 6. Robustness ratio $R = \bar{s}_{\text{pert}}/\bar{s}_{\text{base}}$ decomposed by perturbation axis (5 models $\times$ 3 axes, $n = 1,990$ v2 paired records). Cells centered at $R = 1.0$ (perfect agreement); blue = score drop, red = gain. Model variance dominates axis variance ($3.66\times$), so the robustness column of Table I captures a model property rather than an axis property.

4) *Closed-Form vs Computational Performance*: Phase 2 computational tasks score 0.030 NMACR points higher on average than Phase 1 closed-form tasks (Phase 1 mean 0.6816, Phase 2 mean 0.7115), reversing the “computational is harder” prior. The dimension breakdown explains the inversion: Phase 2 assumption-compliance averages 0.826 versus Phase 1’s 0.420 (+0.41), while numeric accuracy collapses from 0.590 to 0.080 (−0.51). The NMACR weighting ($A = 0.30$ vs $N = 0.10$) lets the assumption gain dominate the numeric loss. Intra-phase rankings flip substantially: Mistral jumps from 5 $\rightarrow$ 2, DeepSeek drops 3 $\rightarrow$ 5, Claude drops 2 $\rightarrow$ 3. The phase axis is a fourth independent rank-divergence dimension beyond accuracy, robustness, and calibration. Phase 2’s zero L1 failure rate (§IV-F2) is consistent with this picture: implicit-assumption sampler prompts produce continuous partial credit on A that the judge accepts and the binary L1 flag does not trigger.

TABLE IV
PER-MODEL NMACR BY PHASE

Model	Phase 1 ($n = 136$)	Phase 2 ($n = 35$)	Δ ($P1 - P2$)	Rank shift ($P1 \rightarrow P2$)
Gemini	0.708	0.824	−0.116	1 $\rightarrow$ 1
Claude	0.698	0.695	+0.003	2 $\rightarrow$ 3
DeepSeek	0.675	0.642	+0.033	3 $\rightarrow$ 5
GPT-4.1	0.673	0.675	−0.002	4 $\rightarrow$ 4
Mistral	0.654	0.722	−0.068	5 $\rightarrow$ 2

Note: Phase 1 = 136 closed-form conjugate tasks; Phase 2 = 35 computational-methods tasks (Gibbs, MH, HMC, RJMCMC, hierarchical, VB, ABC). Δ = Phase 1 − Phase 2 mean NMACR. Rank shift = position change in intra-phase ranking (1 = best). Weighted average $(136P1 + 35P2)/171$ reproduces Table I accuracy within ± 0.001 .

V. ANALYSIS AND VALIDATION

What the three-ranking divergence means. The three metrics are not the same axis under different names. Each has deployment consequences the others do not capture. Accuracy is the natural signal for one-shot question answering. For pipelines that paraphrase prompts or swap inputs (automated tutoring, agentic workflows), robustness matters more: picking by accuracy alone for paraphrase-heavy pipelines can be exactly wrong. For systems that gate decisions on expressed confidence (triage tools, ensemble selectors), calibration matters, and DeepSeek’s ECE of 0.198 rules it out regardless of accuracy.

Reasoning versus pattern matching. The keyword-vs-judge gap on assumption-compliance is the clearest evidence for the paper’s title. On the combined base-and-perturbation pool ($n = 2,850$), 20.7% of responses pass-flip between the two scorers. On the $n = 750$ base-only pool, the judge passes 55.7% versus the keyword rubric’s 71.1%, a 15.4-point gap that indicates the direction: surface-form scoring overcounts substantive articulation [10]. The calibration analysis (Table III) shows the same shape: verbalized confidence places Claude and GPT-4.1 in the well-calibrated tier (ECE ≈ 0.033), while self-consistency on three reruns at $T = 0.7$ puts every model past the ECE = 0.6 “severe overconfidence” threshold.

Broader implications. The multi-axis ranking is cheap to add: the same response corpus feeds robustness once perturbations exist and calibration once confidence is parsed. Single-number leaderboards should be reported alongside the multi-axis view, not in place of it.

Decomposing the divergence: four orthogonal signals. The three-ranking divergence in §IV is one coarse-grained view. §IV-F decomposes it into at least four orthogonal diagnostic signals. (i) Model-specific over-credit (§IV-F1): Mistral $12.11\times$ and Claude $7.84\times$ keyword-judge asymmetry versus GPT-4.1 $2.58\times$ — surface-token credit is not uniform across the benchmarked LLMs. (ii) Task-specific failure concentration (§IV-F2): all 143 L1 failures fall in Phase 1, with Bayesian-regression families (REGRESSION + BAYES_REG) absorbing 35%; failures are uniform across models within each task type, so the over-credit problem and the task-difficulty problem are independent. (iii) Axis-specific robustness (§IV-F3): rephrasing ($R = 0.966$) hurts more than numerical perturbation ($R = 0.978$), with model-variance ($3.66\times$) dominating axis-variance — the robustness ranking is a real model property, but GPT-4.1’s 0.9996 is a near-cancellation of a rephrasing drop and gains on the other two axes. (iv) Phase-specific rank-flips (§IV-F4): Mistral climbs from 5th on Phase 1 to 2nd on Phase 2 (+3 positions), driven by the assumption-compliance bar dropping on procedural prompts where the rubric expects no explicit prior or likelihood statement.

A model can be top-ranked on one signal and bottom-ranked on another — the four are not a single latent factor under different names. Mistral concretely illustrates this: it ranks 5th on Phase 1 NMACR but 2nd on Phase 2; it is the strongest

pattern-matcher per the F.1 asymmetry but the second-most-robust per the F.3 axis decomposition. Pattern matching as a strategy is robust — surface tokens propagate cleanly through paraphrase and number-swap — and it pays off on tasks where the rubric accepts surface-correct procedural prose. It only collapses on tasks with substantive assumption-articulation requirements that the judge actually checks (Phase 1 conjugate derivations, specifically Bayesian linear regression).

A single-metric leaderboard does not just collapse the three-ranking divergence from §IV: it collapses four orthogonal diagnostic axes into one number. The paper’s title question — “Reasoning or Pattern Matching?” — admits a more specific answer than the three-metric view captured: pattern matching is real, model-specific, axis-stable, and rewarded by procedural task structure. A reader picking a model for Bayesian work should inspect at least four columns — assumption-articulation honesty (§IV-F1), task-type failure concentration (§IV-F2), per-axis robustness profile (§IV-F3), and intra-phase rank position (§IV-F4) — alongside the headline accuracy and calibration numbers. The aggregate scoreboard hides which model is genuinely reasoning and which is fluent at the surface.

VI. CONCLUSION

Five frontier LLMs were benchmarked on 171 Bayesian and inferential tasks with 473 perturbations, scored by a keyword rubric, an NMACR aggregator with literature-derived weights, and an out-of-family Llama 3.3 70B Turbo judge constrained to assumption-compliance. The five models produce three different rank orders: GPT-4.1, third on accuracy (0.673), is most robust (0.9996). Claude, second on accuracy (0.698) and top on calibration (ECE 0.033), drops to fourth on robustness (0.9565). 44.3% of responses do not articulate the required assumptions, and the keyword rubric overcounts assumption articulation by 15.4 points relative to the judge. A single number is the wrong instrument for statistical-reasoning evaluation.

Threats to validity. The judge can carry training biases; it is constrained to one dimension and validated against the keyword rubric (Spearman $\rho = 0.602$). The same Llama 3.3 70B Turbo model both generated the v2 perturbations and served as the judge; it evaluates outputs from the five benchmarked LLMs (not Llama outputs), so this is not strict self-judging, but Llama-flavored perturbation phrasing could yield more consistent judge scoring than human-flavored phrasing. Krippendorff’s α sits in the “questionable” range, reflecting genuine ambiguity in scoring free-form Bayesian reasoning. The three-ranking divergence is a within-model finding independent of inter-rater agreement. Task coverage excludes causal inference and experimental design. Model versions are May 2026 snapshots; inversion patterns may not replicate as APIs evolve.

Future work. The four-axis decomposition in §IV-F opens several extensions of this evaluation framework. The task suite can broaden to causal inference, experimental design, and frequentist hypothesis testing, where assumption articulation matters as much as in Bayesian inference. The single-dimension

judge can extend to a multi-judge ensemble with Krippendorff α reported across raters rather than against the keyword rubric only, bounding judge-bias variance more rigorously than the single-rater estimate in §IV-C. The per-model over-credit asymmetry surfaced in §IV-F1 (Mistral 12.11 $\times$, Claude 7.84 $\times$) invites mechanistic investigation: an RLHF artifact, a training-data composition effect, or an attention-pattern signature on assumption-articulation tokens. The Phase 2 zero-failure result in §IV-F2 motivates a rubric redesign that scores procedural assumption articulation — convergence diagnostics, proposal-distribution choice, acceptance-rate calibration — rather than only conjugate-derivation assumptions. The rephrasing-sensitivity result in §IV-F3 (rephrasing damages scores more than numerical perturbation) suggests adversarial perturbation generation targeting the most failure-prone task types from §IV-F2 as a probe corpus for future model releases. Finally, the GPT-4.1 robustness inversion deserves controlled provider-level experiments isolating training-data and decoding choices.

ACKNOWLEDGMENTS

The authors thank Vahe Movsisyan, PhD, and the Akian College of Science & Engineering, American University of Armenia. Claude AI (Anthropic) was used as a writing assistant for grammatical and stylistic refinement.

APPENDIX A

FORMAL METRIC DEFINITIONS

This appendix collects formal definitions for the five quantitative metrics in the main text and the bootstrap procedure used for all reported confidence intervals.

A. NMACR Aggregate

See Eq. (1) in §III-C for the NMACR aggregate definition. The weight ordering $A > R > M > C > N$ reflects assumption articulation as the foundation of Bayesian reasoning; a correct numeric answer without derivation does not support the answer logic. The full literature trail behind the weight values lives in the scoring metadata.

B. Robustness Score Ratio

See Eq. (2). The pool is $n = 398$ LLM-generated v2 perturbation records per model (covering 146 of the 171 base tasks). The 75 hand-authored v1 seeds covering the remaining 25 base tasks are excluded from the robustness ratio to keep the generation source uniform; the v1 seeds remain in the canonical 473-record perturbation superset for other analyses.

C. Expected Calibration Error

See Eq. (3); $B = 4$ confidence buckets are parsed from hedging language (low, medium, high, unstated), with bucket centres 0.25, 0.50, 0.85, 0.50 for computing $\text{conf}(B_b)$.

D. Krippendorff’s α

Inter-rater reliability between keyword rubric and judge is $\alpha = 1 - D_o/D_e$ (D_o observed, D_e chance-expected disagreement), computed on the three Path B dimensions. α is computed on $n = 750$ base paired outputs (150 tasks $\times$ 5 models; the 21 trivial-pass task specifications with empty assumption-check lists are excluded to avoid agreement inflation from auto-PASS responses).

E. Spearman Rank Correlation

Per-dimension agreement is also reported as $\rho = 1 - 6 \sum d_i^2 / [n(n^2 - 1)]$ with d_i the rank difference; ρ is preferred over Pearson r because the rubric-judge relationship is monotonic but not linear. Reported on the same $n = 750$ pool as the Krippendorff coefficient above (see §A-D).

F. Bootstrap Procedure

All CIs use the 95% percentile bootstrap on per-run aggregates, with $B = 10,000$ resamples (seed 42). Resamples preserve per-model count ($n = 171$ accuracy, $n = 398$ robustness).

APPENDIX B FAILURE CASE STUDIES

Three representative L1-failure examples from the $n = 143$ pool (Fig. 3), one from each of the three most-frequent failure categories. Excerpts; full responses are in `data/processed_data/results_v2/error_taxonomy_v2_judge.jsonl` in the public repository.

A. Assumption Violation (Mistral, Bayesian Regression)

Task. BAYES_REG_01: Bayesian linear regression with a Normal-Inverse-Gamma conjugate prior on $n = 10$ observations; report the posterior mean of the intercept and slope.

Response excerpt.

“...We model the data as $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$. In vector form, $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$. The conjugate prior is $\boldsymbol{\beta} \mid \sigma^2 \sim \mathcal{N}(\boldsymbol{\mu}_0, \sigma^2 \boldsymbol{\Lambda}_0^{-1})$, $\sigma^2 \sim \text{Inv-Gamma}(a_0, b_0)$...”

Judge verdict.

“The model does not explicitly state the assumptions of i.i.d., normal errors, and constant variance.”

Interpretation. The response writes out the Normal model and the conjugate prior formally, but never states i.i.d. sampling, normality of errors, or homoskedasticity — precisely the assumptions whose articulation the judge audits. The formal notation embeds the assumptions without naming them, the concrete mechanism behind §IV-F1’s 12.11 $\times$ Mistral over-credit asymmetry.

B. Mathematical Error (GPT-4.1, Box-Muller)

Task. BOX_MULLER_02: Generate two independent standard normal samples from $U = 0.2$, $V = 0.5$ via the Box-Muller transform.

Response excerpt.

“... $-2 \ln(0.2) = 3.2189$, $\sqrt{3.2189} \approx 1.7941$, $2\pi V = \pi$, $\cos(\pi) = -1$, $\sin(\pi) = 0$, so $z_1 = -1.7941$, $z_2 = 0$. ANSWER: $-1.7941, 0.0000$.”

Judge verdict.

“The model correctly applies the Box-Muller formula but the final answer disagrees with the reference; the per-step computation appears to follow the method.”

Interpretation. Procedure-aware response with the right formula, the right intermediate steps, and a clean final-answer format — yet the final numeric result fails the tolerance check. Consistent with §IV-F2’s task-difficulty pattern: failures concentrate by task type rather than by model.

C. Conceptual Error (Gemini, Order Statistic)

Task. ORDER_STAT_02: For $X_1, \dots, X_4$ i.i.d. Uniform[0, 1], evaluate the PDF of $X_{(1)}$ at $y = 0.2$.

Response excerpt.

“... $g_{(1)}(y) = \frac{4!}{0!3!} y^0 (1-y)^3 = 4(1-y)^3$, so $g_{(1)}(0.2) = 4(0.8)^3 = 2.0480$...”

Judge verdict.

“The model incorrectly identified the PDF of the 1st order statistic as $g_1(y) = 4(1-y)^3$, which does not match the given Beta distribution connection.”

Interpretation. Formula application is correct in isolation, but the response misses the prompt’s explicit Beta($k, n-k+1$) connection, defaulting to the textbook order-statistic identity. Less-frequent than B.1 or B.2 (only 10 of 143 cases), but the binary L1 flag captures it on the same footing as the dominant modes.

REFERENCES

- [1] T. Liu, Z. Chen, Z. Fang, W. Luo, M. Tian, and Z. Liu, “MathEval: A comprehensive benchmark for evaluating large language models on mathematical reasoning capabilities,” *Frontiers of Digital Education*, 2025. [Online]. Available: <https://doi.org/10.1007/s44366-025-0053-z>
- [2] C. Nagarkar, L. Bogachev, and S. Sharoff, “Can LLM reasoning be trusted? a comparative study: Using human benchmarking on statistical tasks,” *arXiv preprint*, 2026, arXiv:2601.14479. [Online]. Available: <https://arxiv.org/abs/2601.14479>
- [3] Y. Lu *et al.*, “StatEval: A comprehensive benchmark for large language models in statistics,” *arXiv preprint*, 2025, arXiv:2510.09517. [Online]. Available: <https://arxiv.org/abs/2510.09517>
- [4] R. Longjohn, S. Wu, S. Kher, C. Belém, and P. Smyth, “Bayesian evaluation of large language model behavior,” *arXiv preprint*, 2025, arXiv:2511.10661. [Online]. Available: <https://arxiv.org/abs/2511.10661>
- [5] W. M. Bolstad, *Introduction to Bayesian Statistics*, 2nd ed. Hoboken, NJ, USA: Wiley-Interscience, 2007.
- [6] P. D. Hoff, *A First Course in Bayesian Statistical Methods*, ser. Springer Texts in Statistics. New York, NY, USA: Springer, 2009.
- [7] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.
- [8] Y. Yamauchi, T. Yano, and M. Oyamada, “An empirical study of LLM-as-a-Judge: How design choices impact evaluation reliability,” *arXiv preprint*, 2025, arXiv:2506.13639. [Online]. Available: <https://arxiv.org/abs/2506.13639>
- [9] B. Feuer, C.-Y. Tseng, A. S. Lathe, O. Elachgar, and J. P. Dickerson, “When judgment becomes noise: How design failures in LLM judge benchmarks silently undermine validity,” *arXiv preprint*, 2025, arXiv:2509.20293. [Online]. Available: <https://arxiv.org/abs/2509.20293>
- [10] J. Du *et al.*, “Ice cream doesn’t cause drowning: Benchmarking LLMs against statistical pitfalls in causal inference,” *arXiv preprint*, 2025, arXiv:2505.13770. [Online]. Available: <https://arxiv.org/abs/2505.13770>
- [11] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017. [Online]. Available: <https://arxiv.org/abs/1706.04599>

- [12] E. L. Epstein, J. Winnicki, T. Sornwanee, and R. Dwaraknath, "LLMs are overconfident: Evaluating confidence interval calibration with FermiEval," *arXiv preprint*, 2025, arXiv:2510.26995. [Online]. Available: <https://arxiv.org/abs/2510.26995>
- [13] Y. Wang, S. Ni, Z. Ding, Z. Zhan, Y. Li, and K. Bi, "Evaluating and calibrating LLM confidence on questions with multiple correct answers," *arXiv preprint*, 2026, arXiv:2602.07842. [Online]. Available: <https://arxiv.org/abs/2602.07842>
- [14] L. Zheng *et al.*, "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023. [Online]. Available: <https://arxiv.org/abs/2306.05685>
- [15] A. Romanou *et al.*, "Brittlebench: Quantifying LLM robustness via prompt sensitivity," *arXiv preprint*, 2026, arXiv:2603.13285. [Online]. Available: <https://arxiv.org/abs/2603.13285>
- [16] N. Potamitis, L. Klein, and A. Arora, "ReasonBENCH: Benchmarking the (in)stability of LLM reasoning," *arXiv preprint*, 2025, arXiv:2512.07795. [Online]. Available: <https://arxiv.org/abs/2512.07795>
- [17] J. Wei *et al.*, "Chain-of-thought prompting elicits reasoning in large language models," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022. [Online]. Available: <https://arxiv.org/abs/2201.11903>
- [18] W. Chen, X. Ma, X. Wang, and W. W. Cohen, "Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks," *arXiv preprint*, 2022, arXiv:2211.12588. [Online]. Available: <https://arxiv.org/abs/2211.12588>